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Lesson 9: Neural Networks

Technologies for Artificial Intelligence

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Agenda

- One neuron is one line, and where one line runs out
- A hidden layer: several lines and a vote
- How a network learns: the training loop, and one step worked by hand
- A network in Keras on handwritten digits, and softmax
- Training in practice: memorising, two remedies, deep sigmoid networks, shuffled pixels

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Exercise 8, before we start

- Nothing is collected: we discuss it, ten minutes
- What counts is the **prediction made before plotting**, and its reason
- The common gap: trusting a two-dimensional plot to show every anomaly
- Today: a network’s high training accuracy, alone, says little

What a network adds

- Every boundary so far had its **shape fixed in advance**
- Logistic regression: a line; a tree: boxes; a support vector machine: its kernel’s curve
- Training only chose **where** that shape went
- A neural network learns **the shape itself**
- It is built from one simple part, the unit, repeated in layers: the layers between input and output are **hidden layers**

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Three problems, and a ceiling of 0.97

- **Acceptance test**: 3,000 sensors, pass or fail
- **Two-channel drift**: 800 sensors, correctable or not
- **Digits**: 1,797 handwritten images, 8 × 8 pixels
- The test rig is wrong on **3%** of sensors, so no score can exceed **0.97**
- A hidden layer is worth **39 points** on the sensors, **2.8** on the digits

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The perceptron, and a unit you already have

- The **perceptron** (Rosenblatt, 1958): output 1 if the weighted sum is positive
- Swap that hard step for lesson 4's sigmoid
- The result is one logistic regression unit, `LogisticRegression`
- New today: the unit is a **component**, repeated
- Also called a **unit**, or a **neuron**

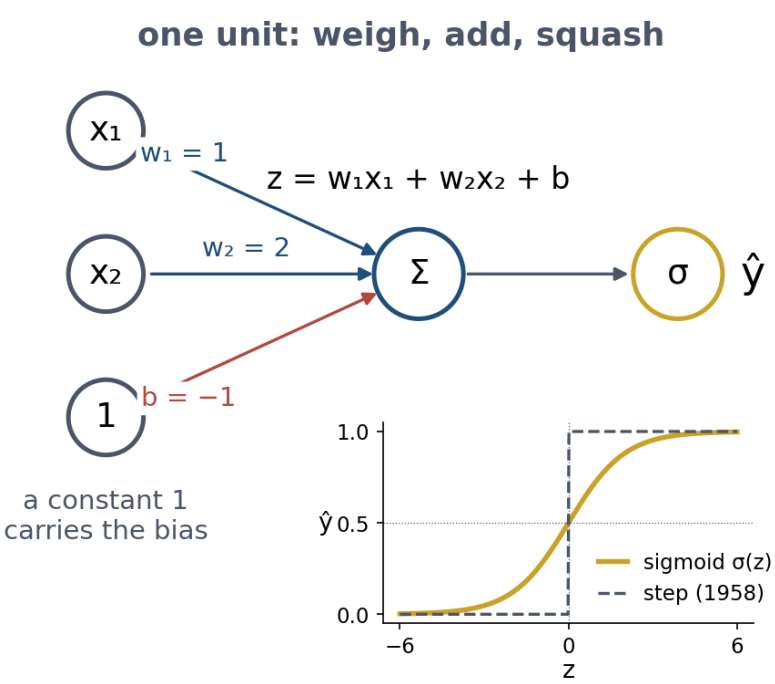
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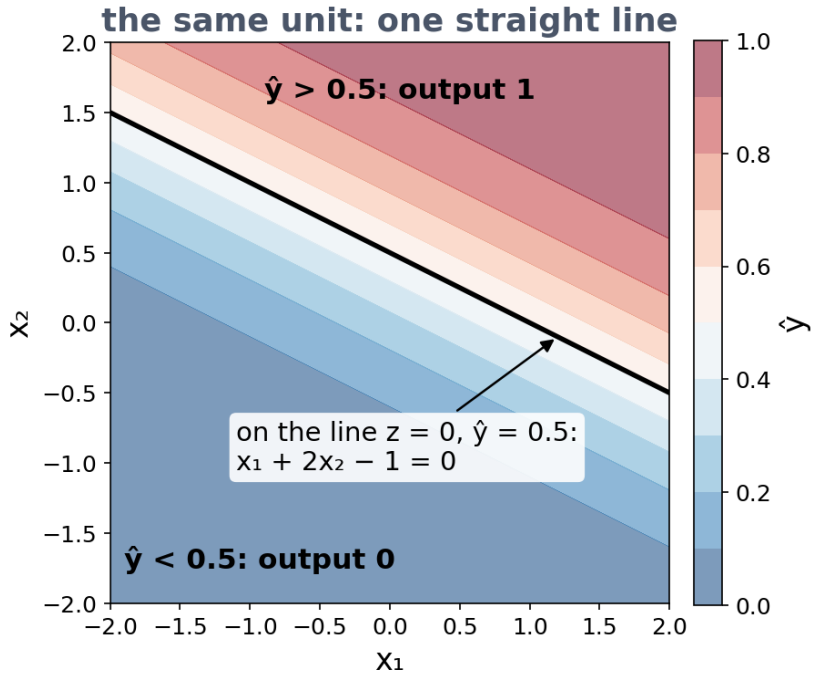
A neuron is a line

$$\hat{y} = \sigma(w^\top x + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}$$

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One neuron, drawn





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One unit, by hand

- Weights $w = (1, 2)$, bias $b = -1$: the unit's line is $x_1 + 2x_2 - 1 = 0$
- Point (1, 1): $z = 1 + 2 - 1 = 2$, and $\hat{y} = \sigma(2) =$ **0.8808**, class 1
- Point (0, 0): $z = -1$, and $\hat{y} =$ **0.2689**, class 0
- Point (1, 0): $z = 0$, and $\hat{y} =$ **0.5000**: on the line, **undecided**
- The unit's verdict is which side of its line a point lies on

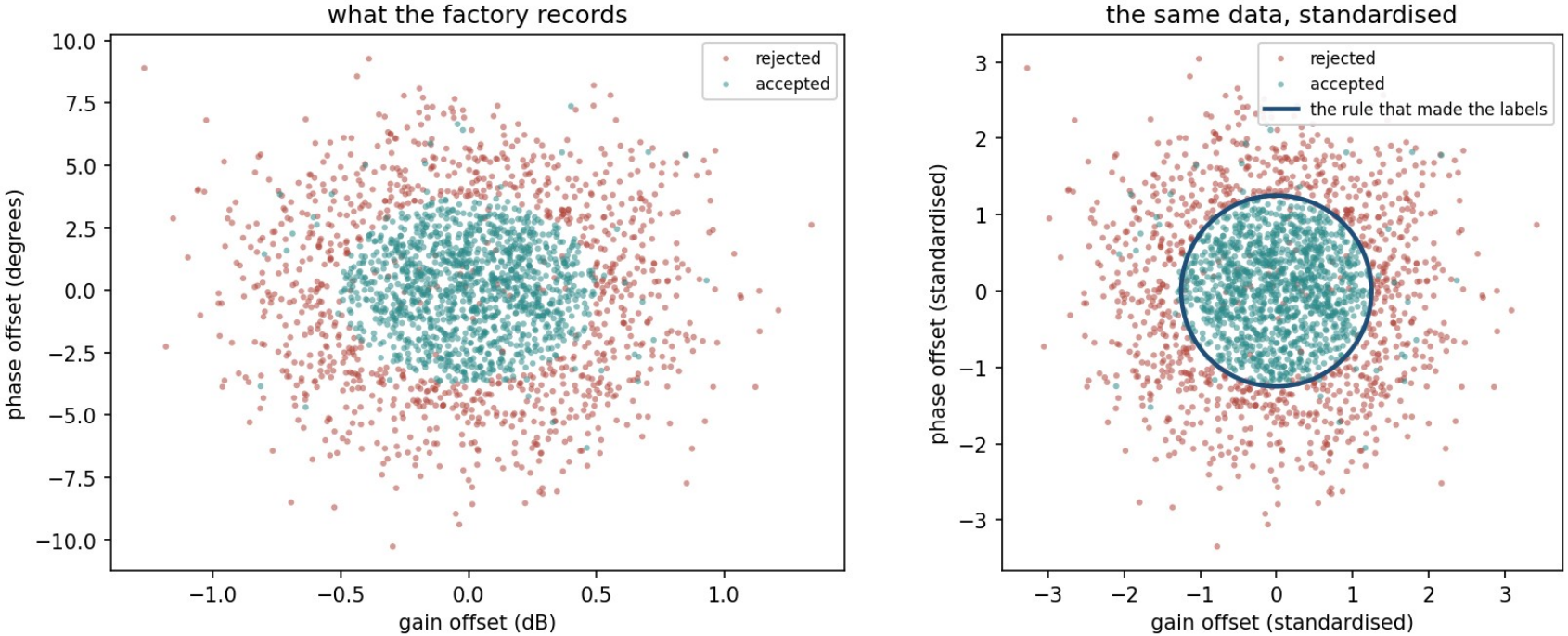
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Meridian’s acceptance test

- Each sensor has a gain offset (decibels) and a phase offset (degrees)
- It passes when both offsets are **jointly** small
- 3,000 sensors: `train_test_split` keeps 2,250 to train and 750 to test
- `StandardScaler`, fitted on the 2,250 only, divides each axis by its spread (standard deviation)
- Both tolerances are 1.25 spreads, so a sensor passes inside a **circle of radius 1.25**

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2,250 sensors, raw and standardised

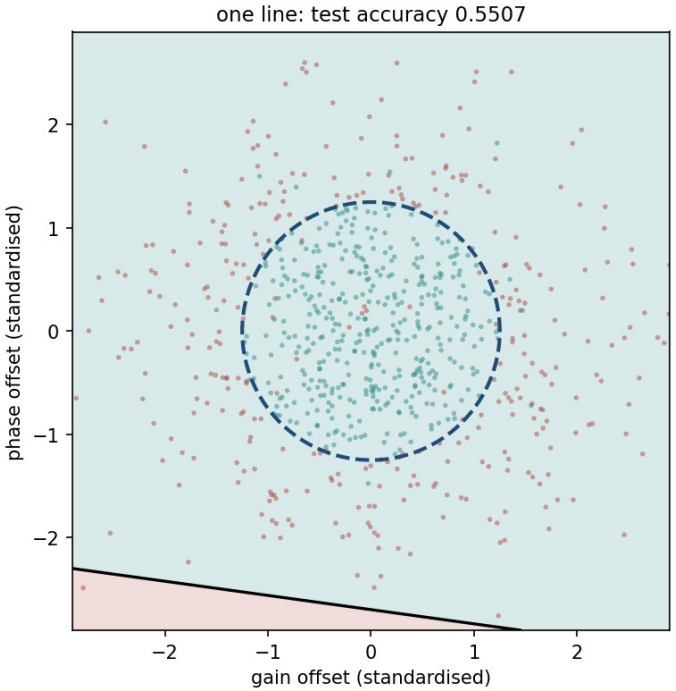


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What one line is worth

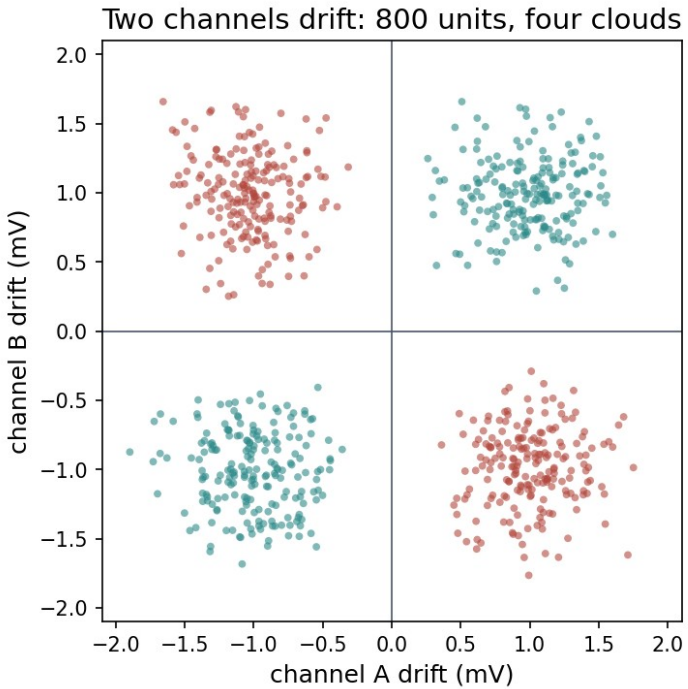
model, scored on the 750 test sensors	accuracy
always predict “accepted”	0.5480
logistic regression: one unit, one line	**0.5507**
the rig’s ceiling	0.9700

No line encloses a region



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New data: two drifting channels



same sign (common mode): correctable
opposite signs (differential): not correctable

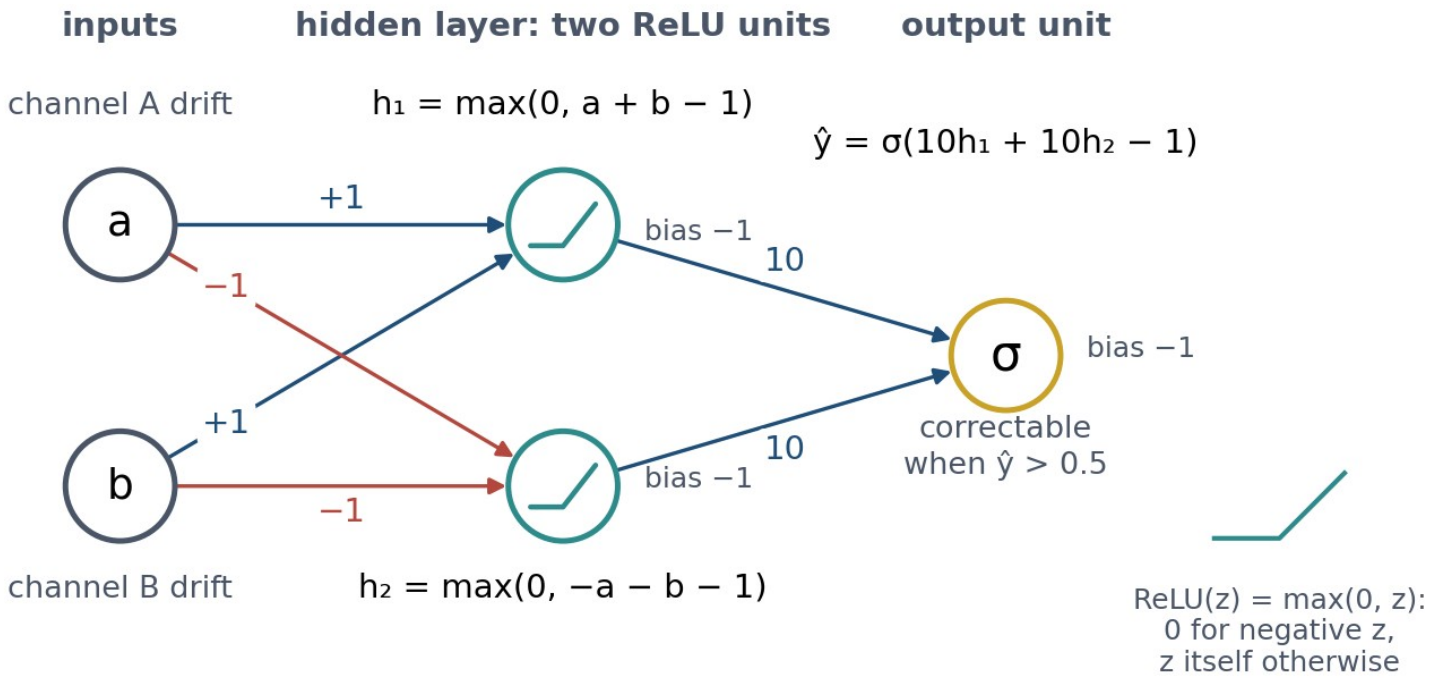
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The smallest problem that needs a hidden layer

- Correctable when the drifts a and b share a **sign**: the exclusive-or (XOR) of the two signs
- Logistic regression, all 800 sensors: exactly **0.5000**
- Correctable clouds have a + b near ±2; the others near 0
- The rule “|a + b| is large” needs **two boundaries**
- One line draws one boundary; two hidden units can draw two

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Two hidden units, drawn

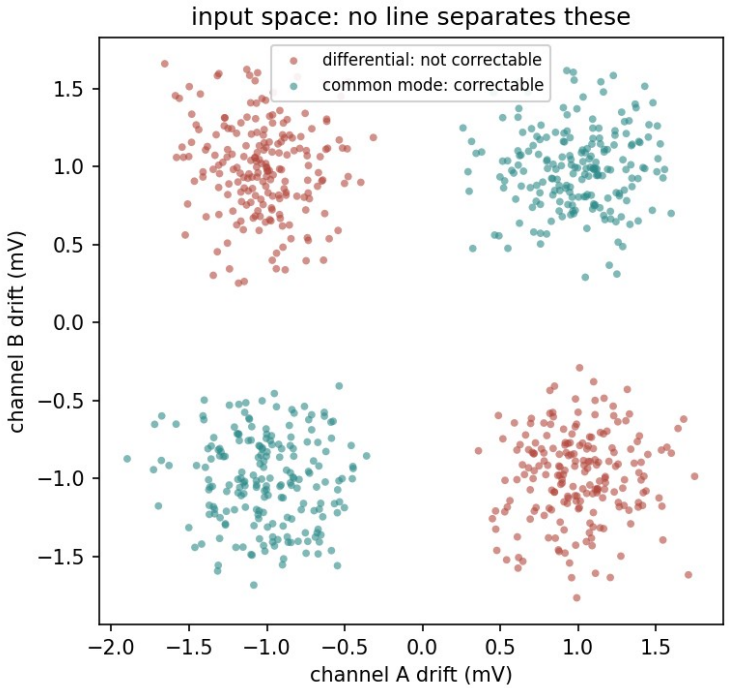


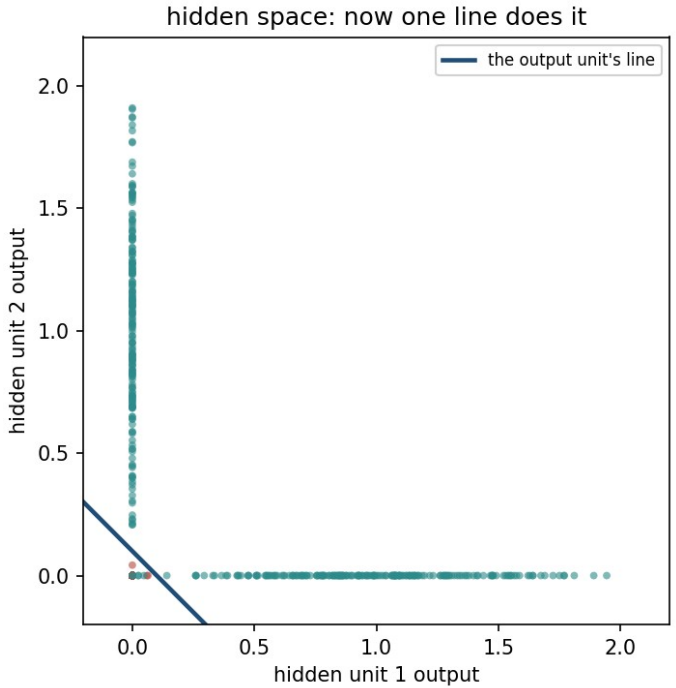
Two units, written down rather than trained

- Each hidden unit applies an **activation** to its sum z: the rectified linear unit (ReLU), max(0, z)
- $h_1 = \max(0, a + b - 1)$ and $h_2 = \max(0, -a - b - 1)$
- Output: $\hat{y} = \sigma(10h_1 + 10h_2 - 1)$
- $\hat{y}$ at centre (+1, +1): 0.9999; at (+1, −1): 0.2689
- All 800 sensors, **nothing trained**: accuracy **0.9938**

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What the hidden layer actually did





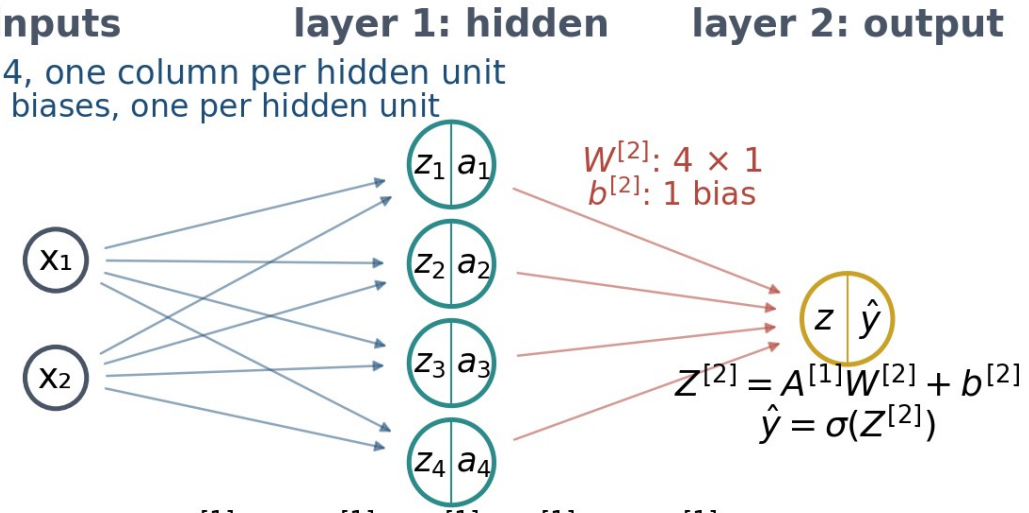
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The hidden layer classified nothing

- The hidden layer **moved the data**
- Correctable clouds went to (1, 0) and (0, 1); the rest to the origin
- There, the output unit’s **single line** is enough
- Learning such new coordinates from data is called **representation learning**
- Computing the output from the input, layer by layer: the **forward pass**

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Forward propagation, drawn



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Forward propagation, written

$$A^{[1]} = g(XW^{[1]} + b^{[1]}), \quad \hat{y} = \sigma(A^{[1]}W^{[2]} + b^{[2]})$$

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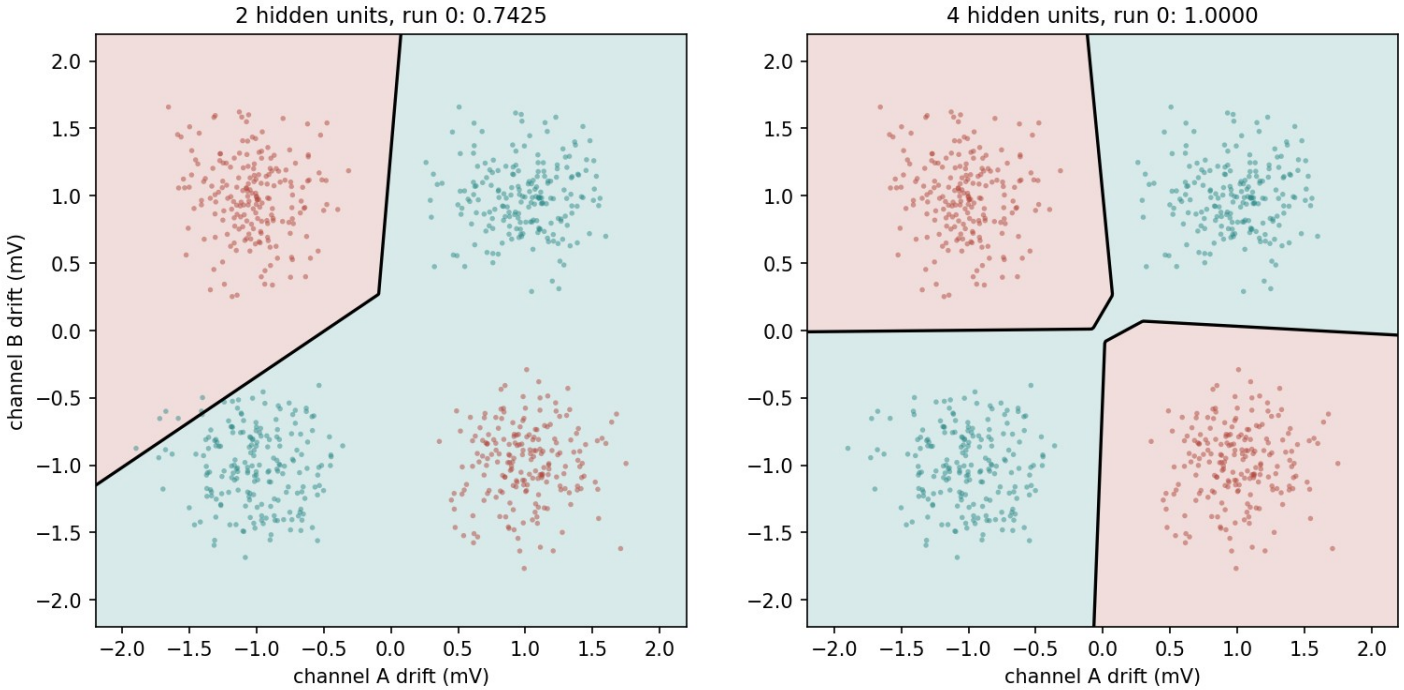
Letting training find the weights

- `MLPClassifier` (multilayer perceptron): one hidden layer, from random weights
- Drift, 800 sensors, 20 runs per size from different `random_state` values
- Runs reaching 0.99 accuracy: **6 of 20** with 2 units, 17 of 20 with 4, all 20 with 8
- In code:

```
MLPClassifier(hidden_layer_sizes=(4,))
```

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Two units can; training often misses them



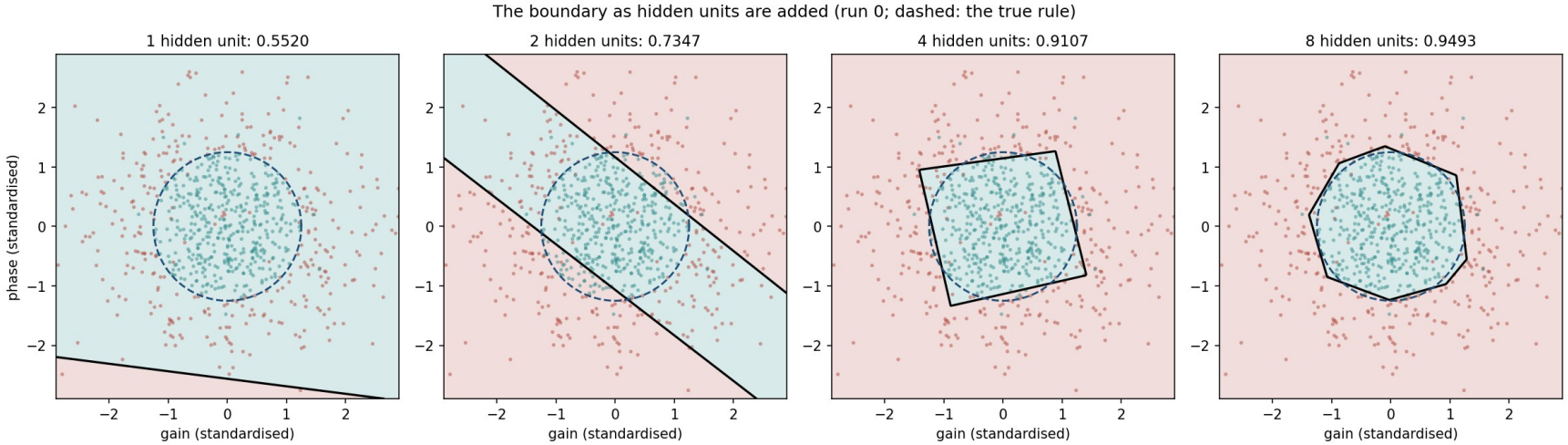
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Acceptance test: how many hidden units?

hidden units (750 test sensors)	mean of 5 runs	worst run	best run
1	0.6048	0.5480	0.6493
2	0.7392	0.7240	0.7693
3	0.8120	0.7227	0.9160
4	0.8821	0.7227	0.9507
8	**0.9445**	0.9387	0.9493
16	0.9549	0.9520	0.9600

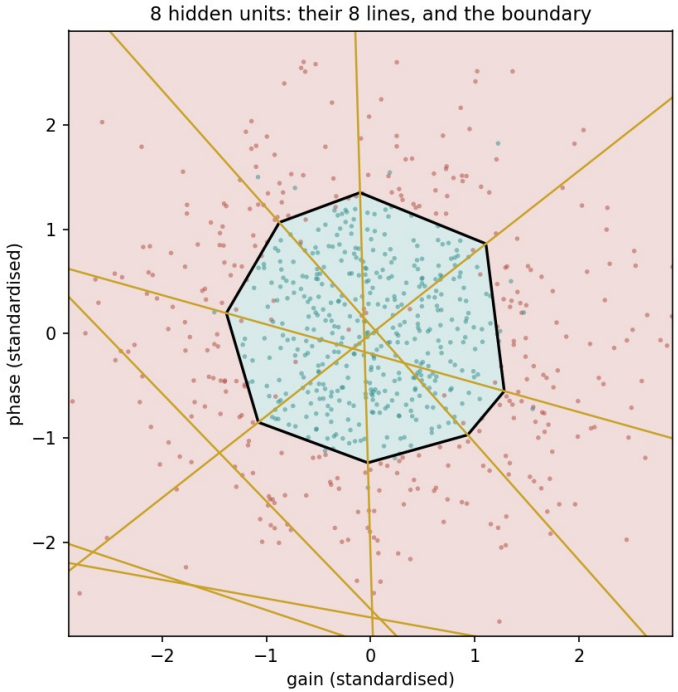
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The boundary at 1, 2, 4 and 8 units



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Each hidden unit is a line



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Notebook 1, live

section, and its slides	you call (→ the number to come back with)
look first, 10-11	`train_test_split`, `StandardScaler` → 2,250 to train, 750 to test
one line, 12-13	`LogisticRegression` → 0.5507 against 0.5480
one unit by hand, 7-9	NumPy only → $\hat{y} = 0.8808$ at (1, 1)
drift, hand-built network, 14-21	NumPy only, the two-unit network → 0.9938
training finds the weights, 22-23	`MLPClassifier`, 20 runs per size → 6 of 20 at two units
widths and hidden lines, 24-26	`MLPClassifier`, then `coefs_[0]` → 0.9445 at eight units
what it was worth, 24	one subtraction → 39.4 points

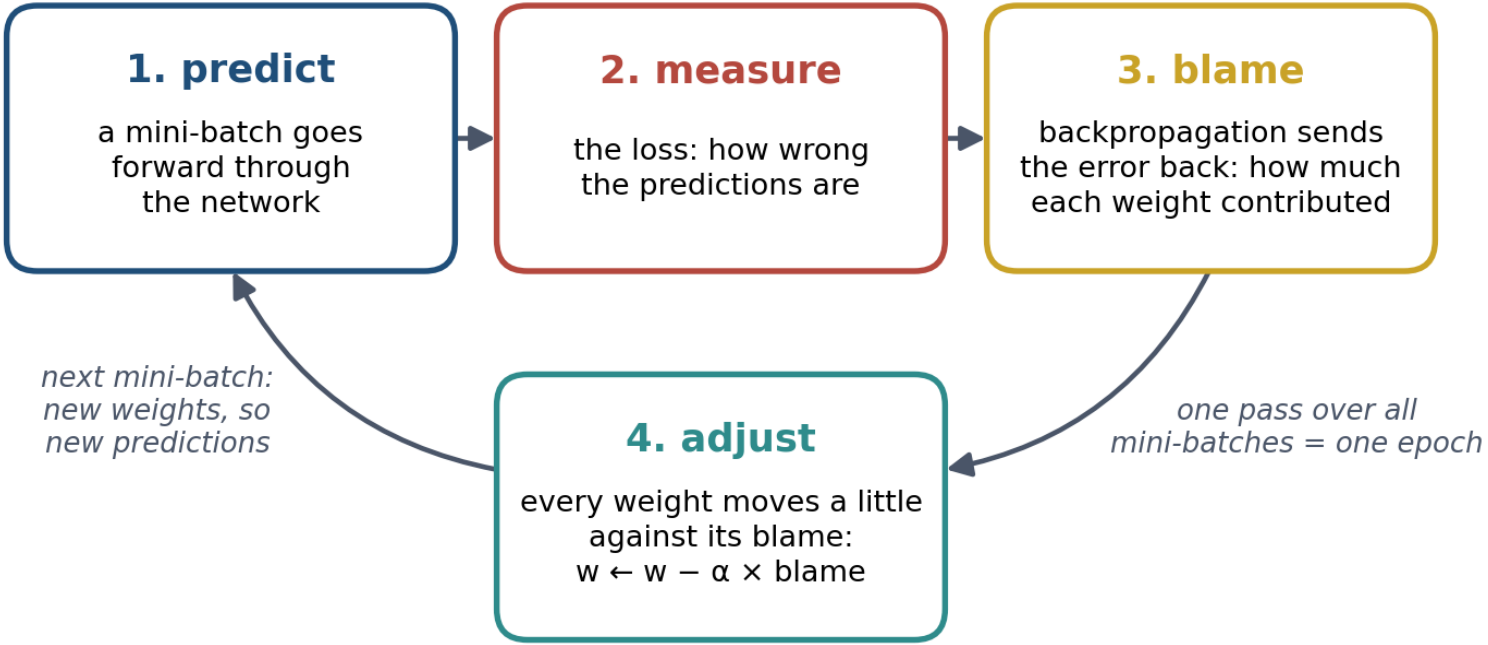
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Break

- Twelve minutes

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How a network learns: the loop



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Mini-batches and epochs

- A **mini-batch**: a small random group of training examples, 32 in notebook 2
- Each round of the loop uses one mini-batch
- An **epoch**: one pass through all the mini-batches
- Digits: 1,077 training images in batches of 32 make **34 per epoch**
- 60 epochs make **2,040 rounds** of the loop

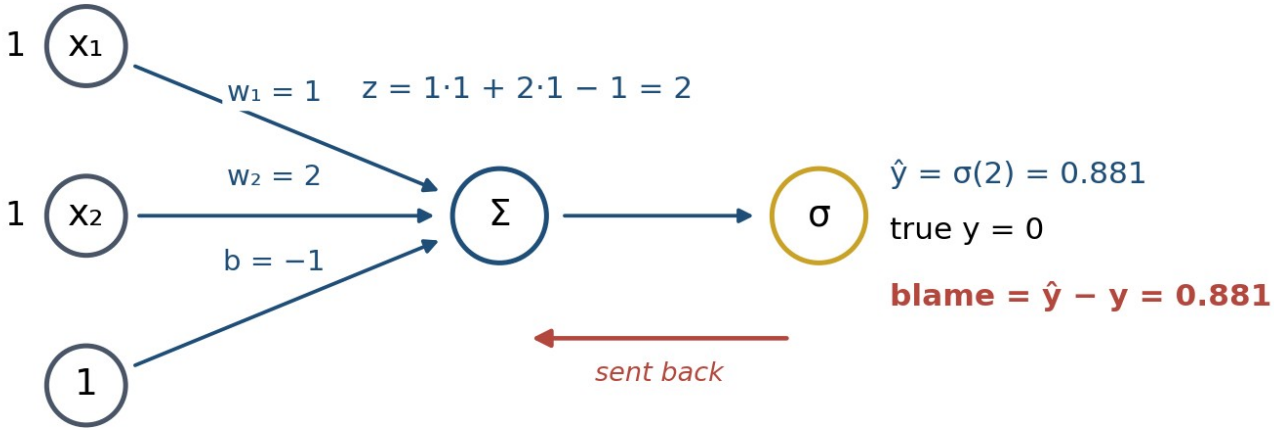
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The loss and the gradient step

- Each weight moves against its **gradient**: how fast the loss (lesson 4's cross-entropy) changes as that weight changes
- $w \leftarrow w - \alpha \cdot \partial L/\partial w$, with α the **learning rate** (lesson 3)
- A weight's gradient is its **blame**: its share of the error
- One sigmoid unit: blame $\hat{\mathbf{y}} - \mathbf{y}$ at the output; each weight's blame is $\hat{y} - y$ times its input

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One step on one unit



each weight's blame = 0.881 × its input; step with α = 0.5:

$w_1 \to 0.560$, $w_2 \to 1.560$, $b \to -1.440$

new prediction: $\hat{y} = \sigma(0.679) = 0.663$, closer to the true 0

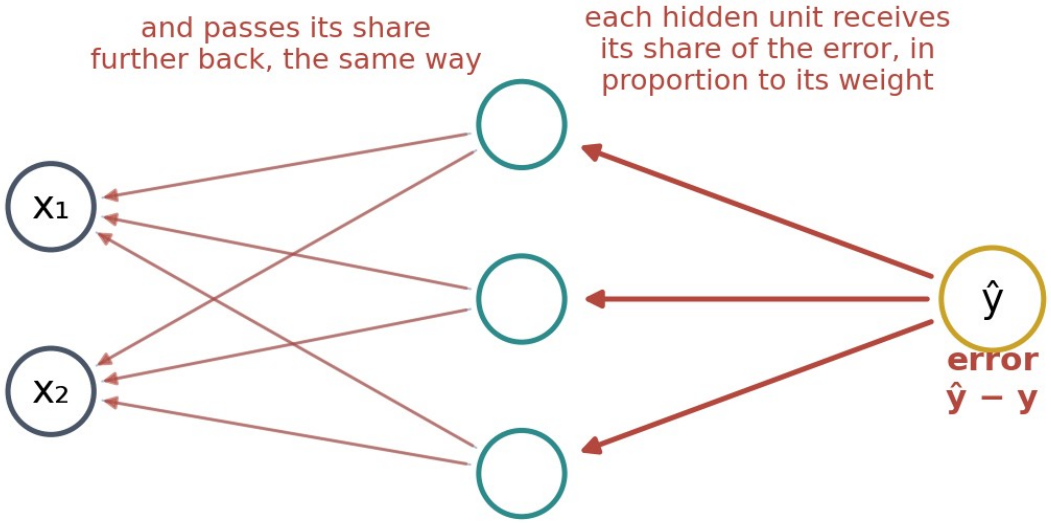
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Reading the step

- Unit $w = (1, 2)$, $b = -1$; input $x = (1, 1)$, true label $y = 0$
- Forward: $z = 2$, $\hat{y} = 0.881$, loss **2.127**
- Blame $\hat{y} - y = 0.881$ for every weight, since both inputs are 1
- Step with $\alpha = 0.5$: each weight down by 0.440, to 0.5596, 1.5596 and −1.4404
- New $\hat{y} =$ **0.663**, loss down to **1.089**

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Backpropagation: blame flows back



one backward sweep gives every weight its blame, layer by layer, right to left

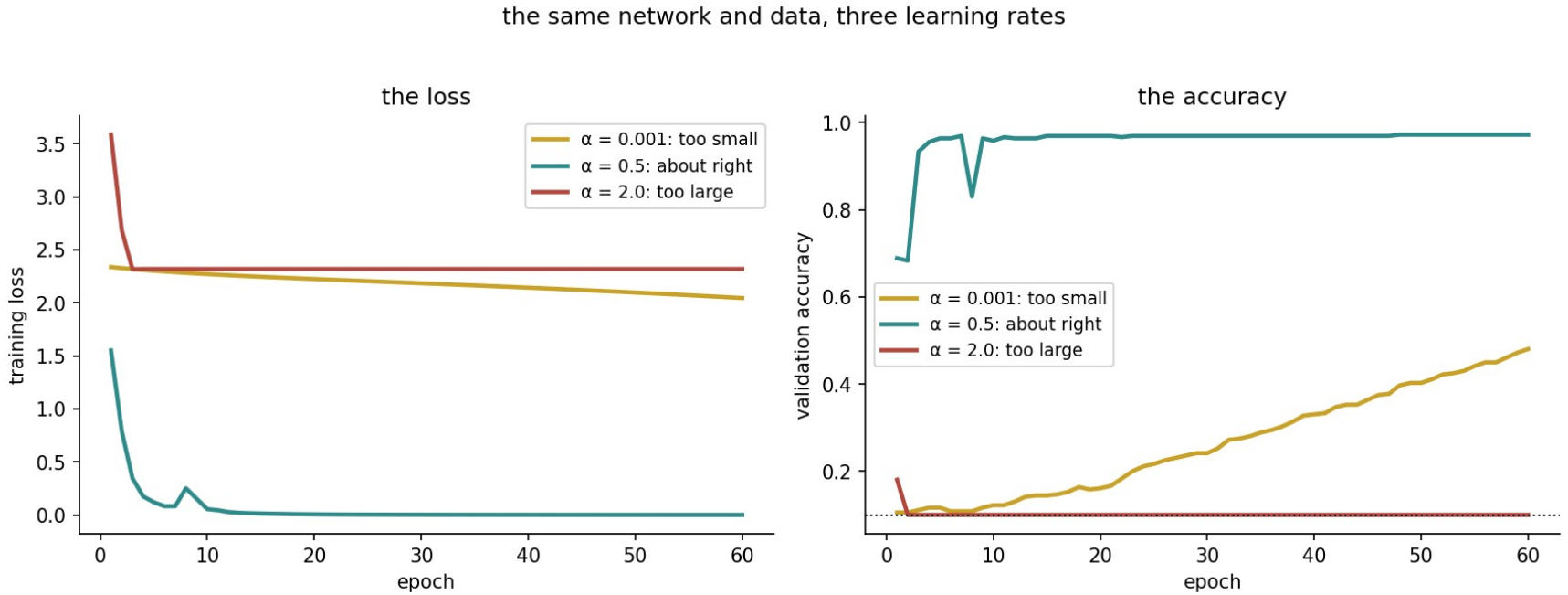
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What backpropagation buys

- **Backpropagation**: the chain rule, applied from the output back
- Each hidden unit receives a share of the output’s error, in proportion to its weight
- It multiplies that share by its activation’s slope, then passes it back
- One backward sweep costs about one forward pass
- scikit-learn and Keras run it inside `fit`

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The learning rate, three regimes



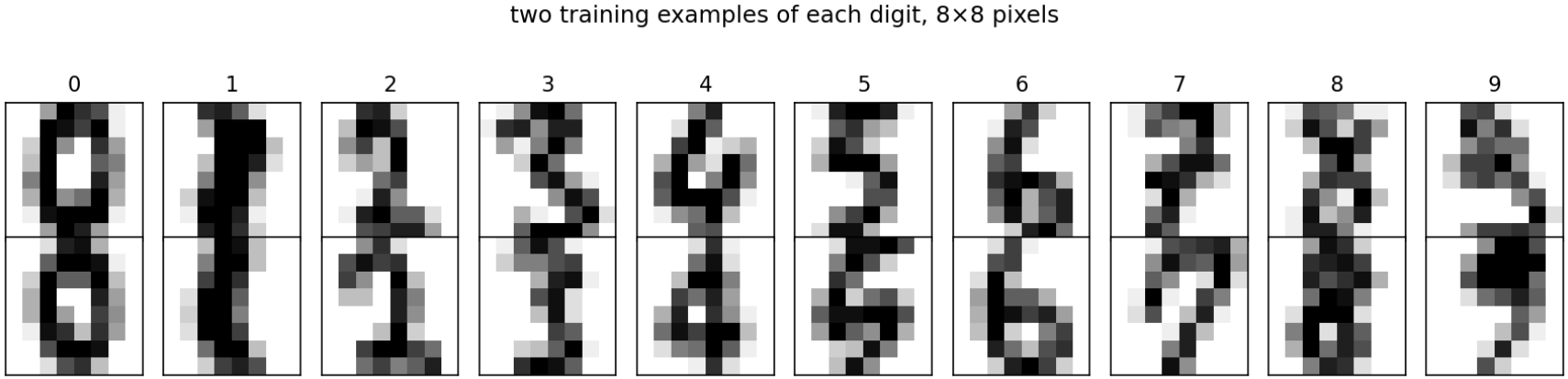
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Predictable mistake: a rate too small

- Two layers of 64, stochastic gradient descent (SGD: one step per mini-batch), 60 epochs, one run per rate
- Validation digits: 0.4806 at $\alpha = 0.001$, **0.9722** at 0.5, 0.1000 at 2.0
- A rate too small looks like **a network too small**
- **Try rates a factor of ten apart** first
- In code: `keras.optimizers.SGD(0.5)`

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Ten digits, eight by eight pixels



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A network in Keras: 2,410 parameters

- **Keras**: the network library of lessons 9 and 10
- `keras.Sequential`: the layers in order, `keras.Input` then two `layers.Dense`
- A **dense** layer: every unit sees every input (32 ReLU, then 10 softmax)
- `compile` picks `Adam` (a refined gradient descent); `fit` trains
- Parameters: 64 × 32 + 32 = 2,080, plus 32 × 10 + 10 = 330

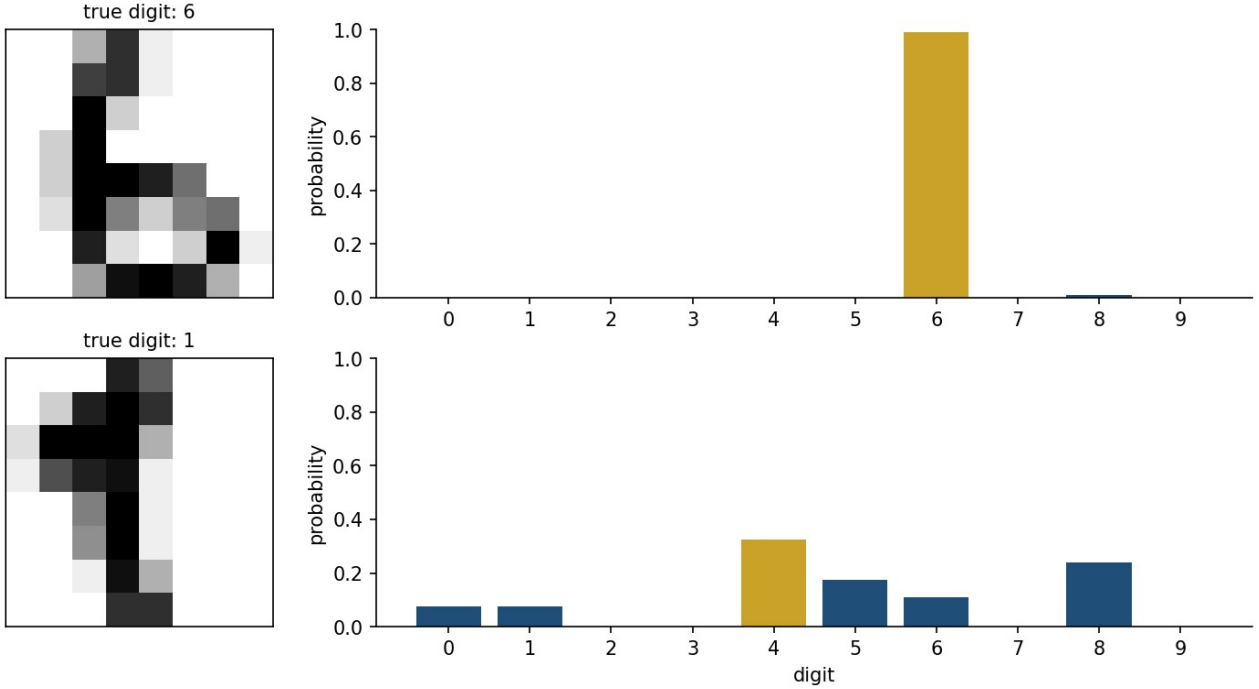
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Softmax

$$\text{softmax}(z)_k = \frac{e^{z_k}}{\sum_{j=1}^{10} e^{z_j}}$$

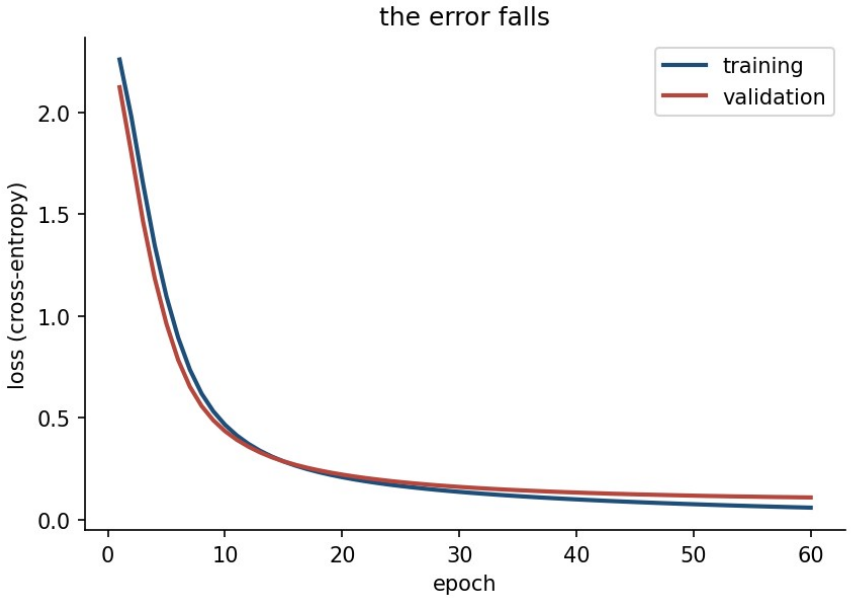
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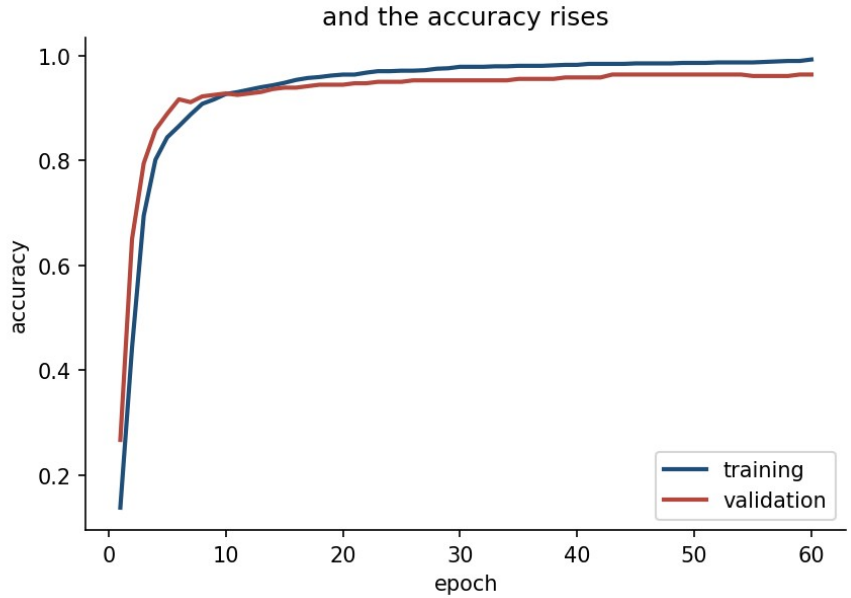
Ten scores become ten probabilities



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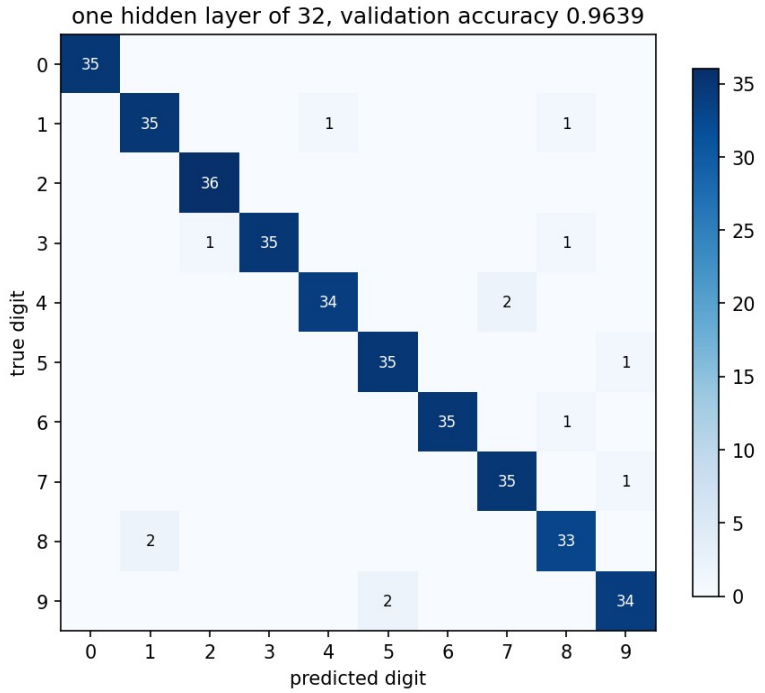
Training, watched





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Where it goes wrong



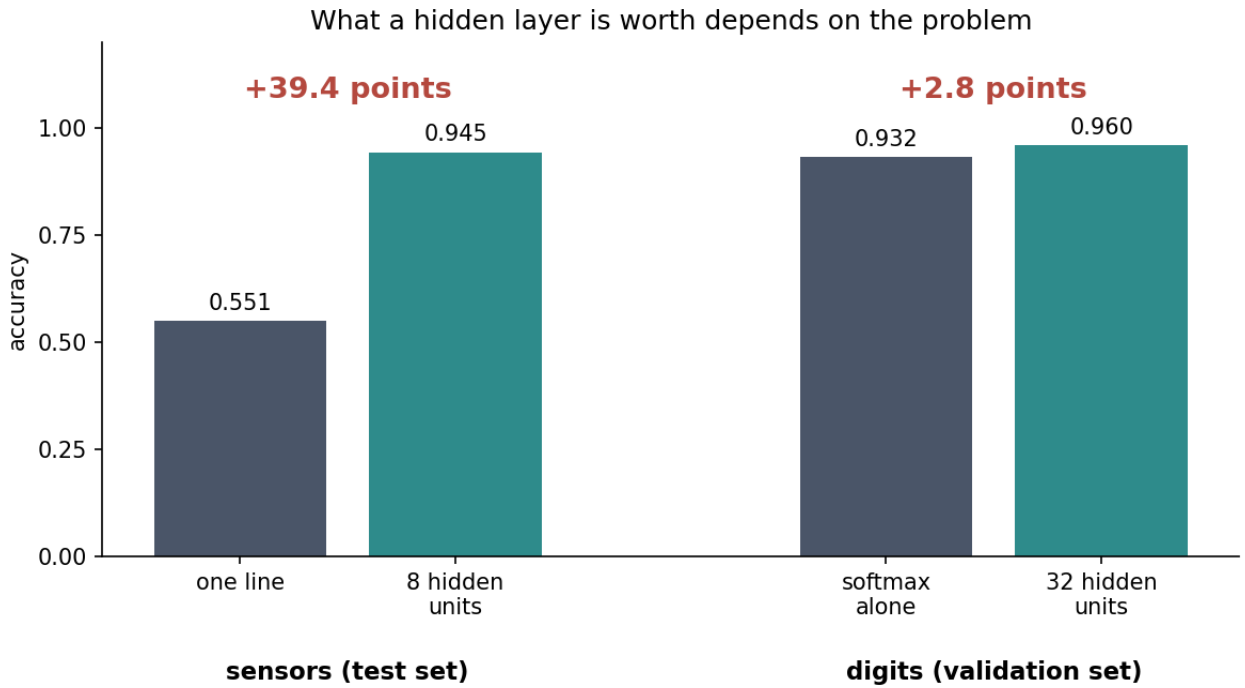
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Is the hidden layer worth it here?

architecture, on the 360 validation digits	parameters	accuracy, mean of 3 runs	standard deviation, 3 runs
softmax alone, no hidden layer	650	0.9324	0.0035
one hidden layer of 32	2,410	**0.9602**	0.0052
one hidden layer of 64	4,810	0.9611	0.0045

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39 points against 2.8



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When to reach for a hidden layer

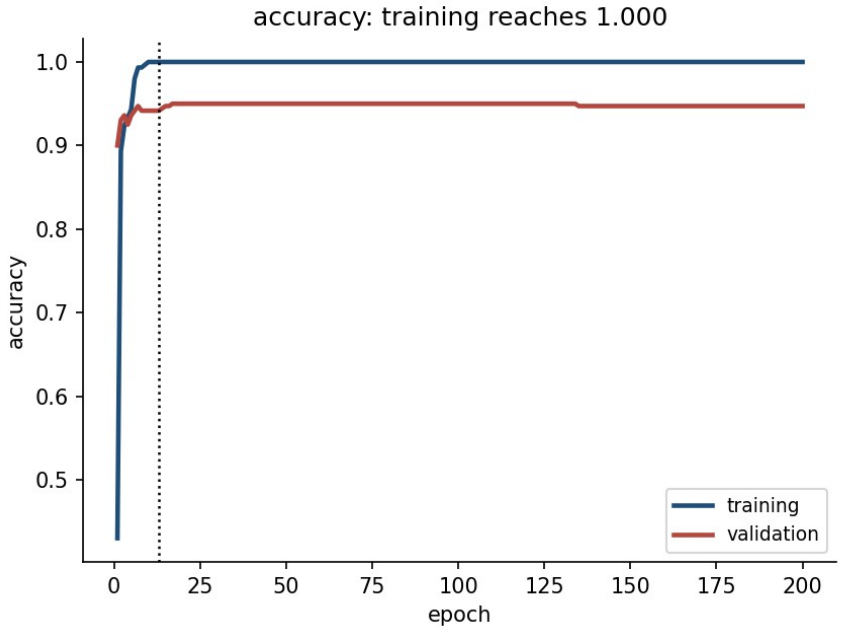
- Sensors: the two offsets act **together**
- Digits: each pixel pushes **on its own**
- The predictable mistake: adding a hidden layer every time, **by reflex**
- Digits: 2,410 parameters against 650, for 2.8 points
- **Fit the linear model first**: its gap to a network says whether a network is needed

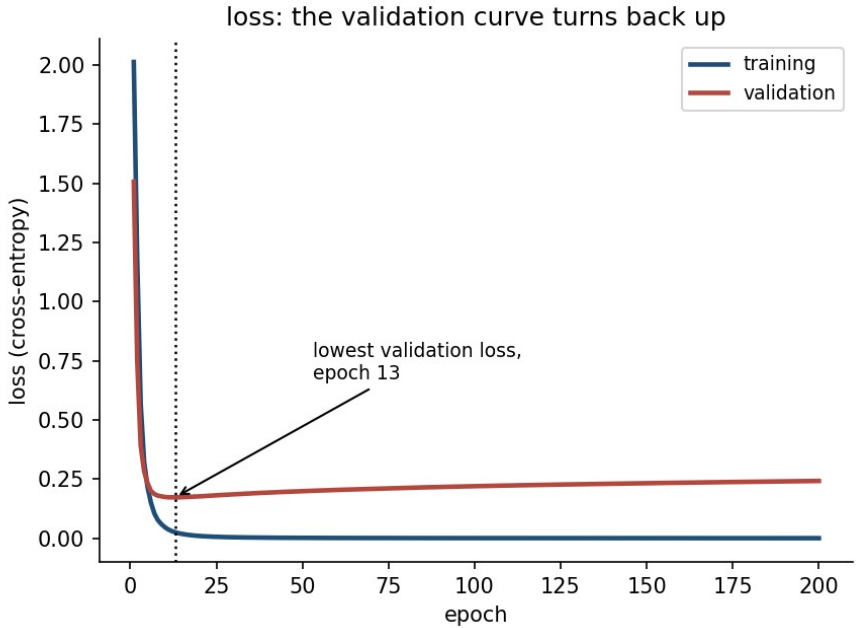
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Notebook 2, live

section, and its slides	you call (→ the number to come back with)
the digits, 38	`train_test_split`, twice → 1,077, 360 and 360 images
building and training, 29-30, 39, 42	`keras.Sequential`, `keras.Input`, `layers.Dense`, `Adam`, `fit` → 2,410 parameters, 0.9639
softmax and confusion matrix, 40-43	`model.predict`, `confusion_matrix` → 13 of 360 wrong
is the hidden layer worth it, 44	three architectures, 3 runs each → 0.9324, 0.9602, 0.9611
the number, 45-46	`LogisticRegression`, `MLPClassifier` on the sensors again → 39.4 and 2.8 points; 4.7 on another split

A network that memorises





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Reading the two curves

- One run: two layers of 512 on **300 digits**: 301,066 parameters
- Training accuracy **1.000**: the 300 answers are stored
- Validation loss lowest at **epoch 13**, 1.4 times that by epoch 200
- Validation accuracy stays near 0.95 (0.9472 at epoch 200), while the network grows more confident about the answers it gets wrong

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Two remedies

- **Early stopping**: stop after 25 epochs with no better validation loss; keep the best weights
- **Dropout**: switch off a random 40% of hidden units on each training mini-batch
- In code: `EarlyStopping(patience=25, restore_best_weights=True)`
- In code: `layers.Dropout(0.4)`

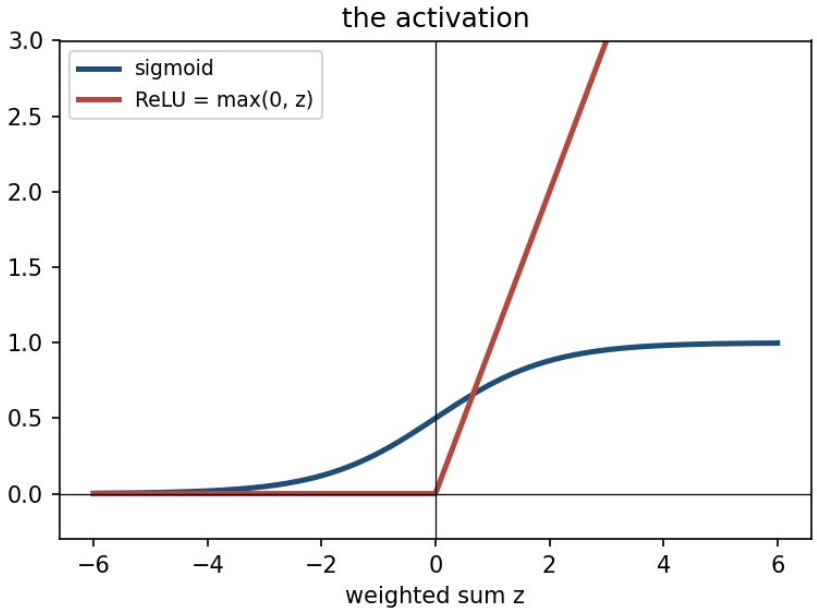
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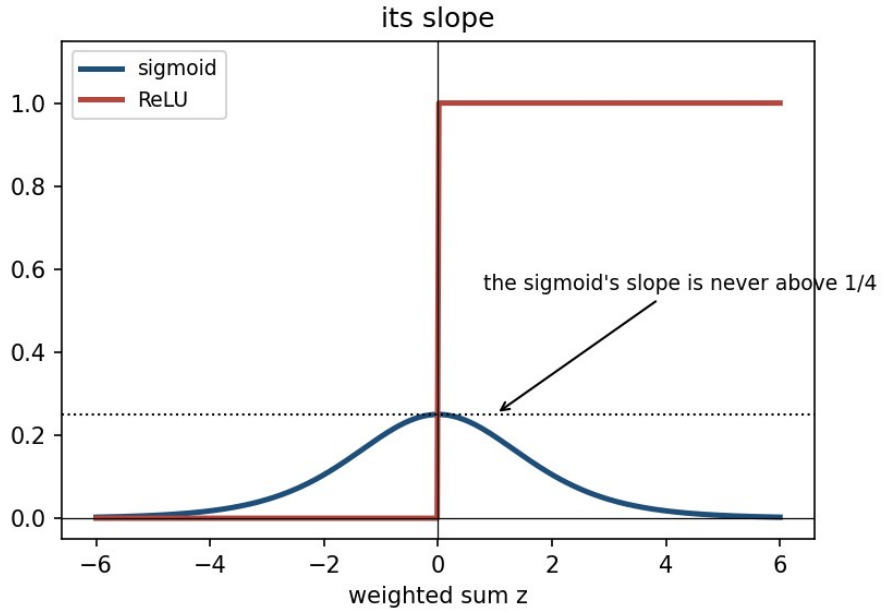
Two remedies, measured

method, two layers of 512 on 300 digits	test accuracy, mean of 3 runs	standard deviation, 3 runs	epochs run
200 epochs, no remedy	0.9426	0.0013	200
early stopping	0.9463	0.0035	39
early stopping + dropout 0.4	0.9481	0.0035	50.3

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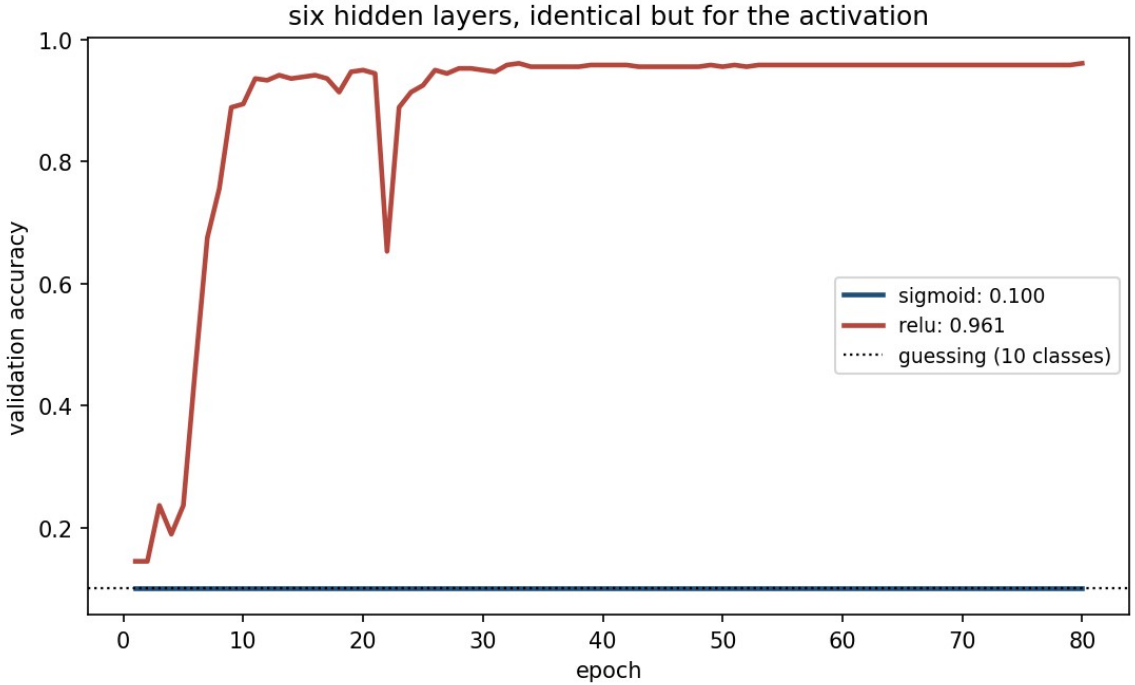
Activations and their slopes





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Six sigmoid layers never learn



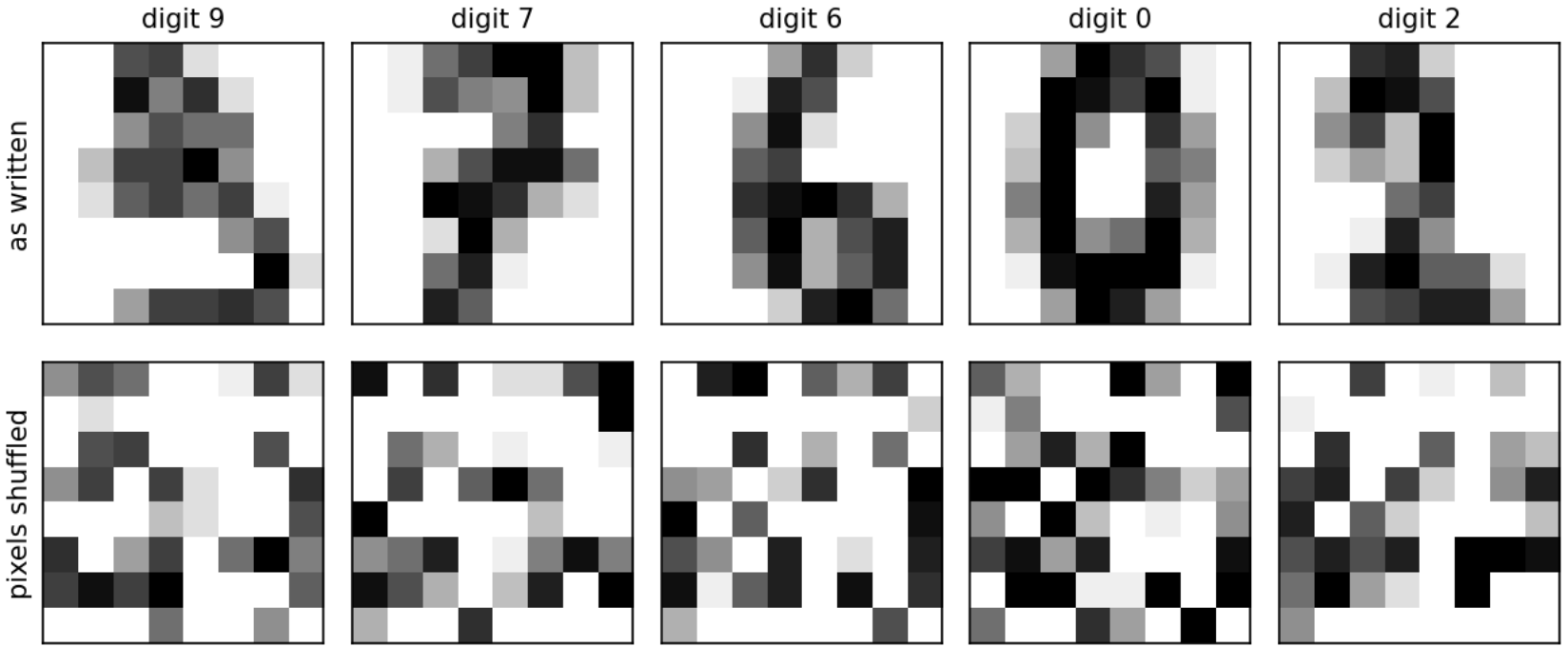
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The vanishing gradient

- Each layer multiplies the blame by its **slope**
- Slope at most 1/4 for the sigmoid; 1 for an active ReLU
- Six sigmoid layers: at most $0.25^6 = 1/4096$ of the blame reaches the first
- Six layers of 32, one run: sigmoid **0.1000**, ReLU **0.9611**
- This is the **vanishing gradient**

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Shuffle the pixels



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Nothing notices

- One fixed reordering of the 64 pixels
- Hidden layer of 32, 3 runs: **0.9602** as written, **0.9630** shuffled
- A difference well inside the run-to-run spread
- A dense layer never uses which pixel is next to which
- **Permute the pixels consistently and nothing here notices.**

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Notebook 3, live

section, and its slides	you call (→ the number to come back with)
the learning rate, 36-37	`keras.optimizers.SGD` at three rates → 0.9722 at α = 0.5
memorising, 48-49	two `Dense` layers of 512 on 300 digits → validation loss lowest at epoch 13
two remedies, 50-51	`EarlyStopping`, `Dropout(0.4)`, 3 runs each → 0.9426, 0.9463, 0.9481
sigmoid against ReLU, 52-54	six `Dense` layers of 32 → 0.1000 against 0.9611
shuffled pixels, 55-56	one fixed permutation of the 64 pixels → 0.9602 against 0.9630

What to take away

- **One neuron is one line**; a **hidden layer moves the data**
- **Learning is a loop**: predict, measure the loss, share out the blame, adjust the weights
- Low accuracy? First try learning rates ten times apart
- **Fit the linear model first**: its gap to a network says if one is needed

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39 points against 2.8

- One hidden layer on the acceptance test: **+39.4 points** (8 units, mean of 5 runs, 750 test sensors)
- One hidden layer on the digits: **+2.8 points** (32 units, mean of 3 runs, 360 validation images; 4.7 on another split)
- On the sensors the offsets act **together**; on the digits each pixel acts mostly **on its own**
- What a network is worth depends on the **shape of the problem**

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Homework

- **Exercise 9**, discussed at the start of lesson 10, **Friday 4 December**
- `Exercises/09_neural_networks.md`
- Two batches from Meridian’s two final-test stations: on which one is a hidden layer worth it, and why?
- Next week starts from today’s last finding: permute the pixels consistently and nothing here notices

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